

EDUCATION

Carnegie Mellon University <i>M.S. in Software Engineering</i> GPA: 4.0 / 4.0 Selected Coursework: Machine Learning, Large Language Model Systems, Data Science for Software Engineering	Expected May 2027
University of Wisconsin–Madison <i>B.S. in Computer Science & B.S. in Computer Engineering</i> GPA: 3.72 / 4.0 Selected Coursework: Large Language Models in Practice, Algorithms, Operating Systems	May 2024

WORK EXPERIENCE

NewsBreak <i>AI Engineer Intern</i>	May – August 2026 <i>Mountain View, CA</i>
- Shipped an agentic LLM search API: deterministic RAG on the first turn, a bounded tool-calling agent on follow-ups (≤ 3 hops, 5 tools), hybrid retrieval fusing full-text and bge-embedding vector search with RRF, SSE streaming, and server-resolved citations the model cannot author. - Designed and shipped a personalized LLM news-alert pipeline as a Kubernetes CronJob (ingest $\rightarrow$ embed $\rightarrow$ LLM-score $\rightarrow$ LLM same-story dedupe $\rightarrow$ push); evaluated it on 4,100+ production shadow decisions and hand-graded alerts, cutting filler 34% $\rightarrow$ 4%. - Took real logged queries that previously returned nothing from 0% to 100% P@1 with a deterministic entity resolver as the query rewriter, after measuring embeddings against it on an eval harness run on the real catalog. - Built the production Python FastAPI / MongoDB backend serving these features from its first commit to 89 REST endpoints on AWS EKS (Docker, Helm), with $\sim 1,450$ offline tests behind a ruff / pyright / pytest gate. - Diagnosed a 65% failure rate in another team’s LLM pipeline, via Langfuse traces, as an unbounded reasoning loop hitting a 16,384-token ceiling; sized a 2,048-token cap against observed p95 of 1,197.	
Haddee Education <i>Engineering Lead</i>	June – September 2025 <i>San Diego, CA</i>
Led an 8-intern team to ship a React, Next.js, FastAPI, and Supabase web app on GCP App Engine; designed schemas and REST APIs for LangChain LLM resume parsing, suggestions, and AI job matching.	
ZenAI — MiraclePlus (Formerly YC China) 2023 Fall Batch <i>Full Stack Developer</i>	September – December 2023 <i>Beijing, China</i>
- Built a VC-backed startup’s LLM mock-interview web app from 0$\rightarrow$1 as the sole engineer (React, Next.js, Supabase, PostgreSQL); owned authentication, session state, and database interactions. - Worked with recruiters so AI scoring categories matched real hiring and iterated scoring and UI from user-testing sessions; demoed the product and obtained contracts with mid-sized companies.	

RESEARCH EXPERIENCE

Control & Learning Group — Carnegie Mellon University <i>Graduate Researcher</i> Advisor: Prof. Yorie Nakahira	February – June 2026
Mapped natural-language instructions to formal safety specs with multi-turn LLMs; built an adaptive probabilistic safety certificate with PhD researchers that outperformed Adaptive MPC in simulation.	

SKILLS

Languages	Python, SQL, Java, Go, TypeScript, JavaScript, C
AI & ML	RAG, bge embeddings, hybrid retrieval, RRF, LangChain, Langfuse, PyTorch, NumPy
Cloud & Infra	Kubernetes (EKS), Docker, Helm, AWS, GCP, CronJobs, GitHub Actions, CI/CD, Linux
APIs & Data	FastAPI, REST APIs, Pydantic, MongoDB, PostgreSQL, MySQL, Supabase
Testing & Practices	pytest, Unit Testing, TDD, ruff, pyright, Git, Agile/Scrum, Jira
Certifications	AWS Certified Solutions Architect – Associate, AWS Certified Cloud Practitioner

PROJECTS & LEADERSHIP

- my-av:** Built a PyTorch CNN-GRU that predicts steering angle and speed from 20-frame (1s) video; built a data pipeline over 33 hours of Comma2k19 and ran streaming inference on local CUDA and GCP Vertex AI.
- ECE Leadership & Engagement Fellow** (CMU, Aug 2026 – present): Own budgeting, marketing, and outreach for ECE student programs — company-sponsored hackathons, alumni events, and tech event outings.